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Snow in the Earth System & Snow cover in Arctic vegetation monitoring

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SIOS RSTC : Satellite and UAV Remote Sensing of the Arctic Biosphere

16.09.2026

Photo credit : Pixabay /
enriquelopezgarre

Why does snow matter?





"Seasonal snow cover is the largest single component of the cryosphere, covering 50% of the northern hemisphere's land surface during mid-winter, and is an important component of the climate system."

climate.esa.int/en/projects/snow

01-2004 MODIS TCC

Slide credit : Kristoffer Aalstad (UiO)

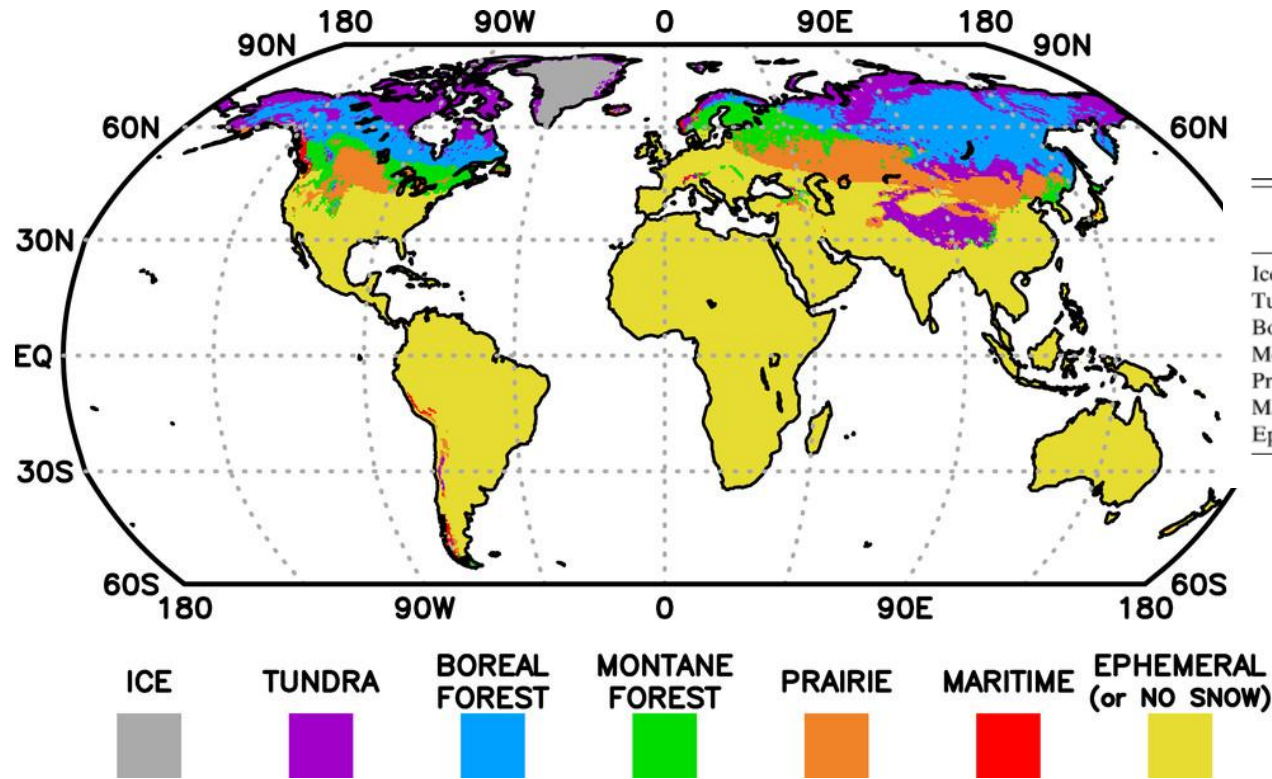


Snow : the Earth's white blanket

08-2004 MODIS TCC

Slide credit : Kristoffer Aalstad (UiO)

Global snow classification



Snow class	Area (10 ⁶ km ²)	Percent Coverage (%)
Ice	14.6	—
Tundra	13.7	31.8
Boreal Forest	12.2	28.3
Montane Forest	6.0	13.8
Prairie	10.5	24.4
Maritime	0.7	1.7
Ephemeral (or No Snow)	89.5	—

Snow : the Earth's reflective white blanket

“Without snow cover, the ground absorbs about four to six times more of the sun's energy” - NSIDC

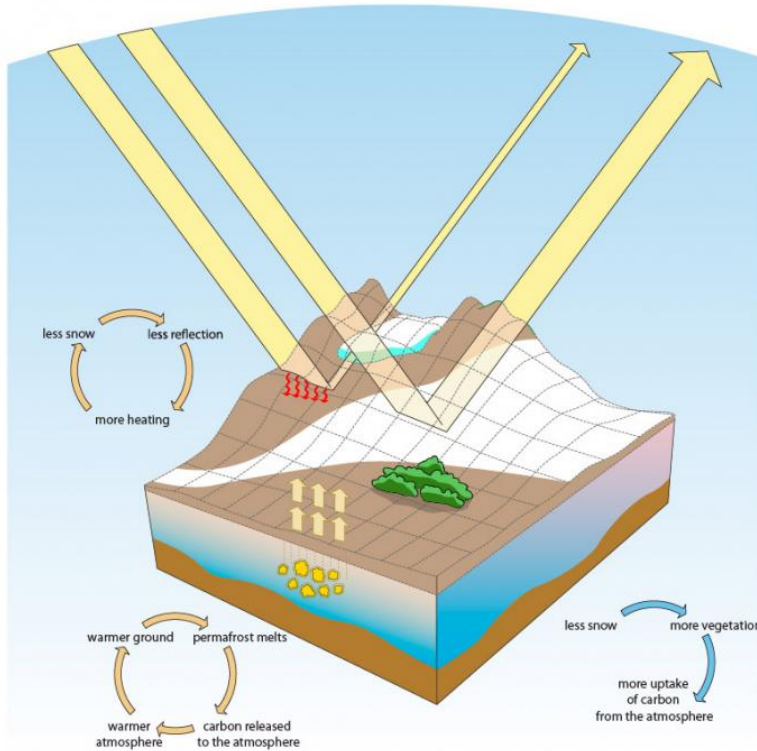


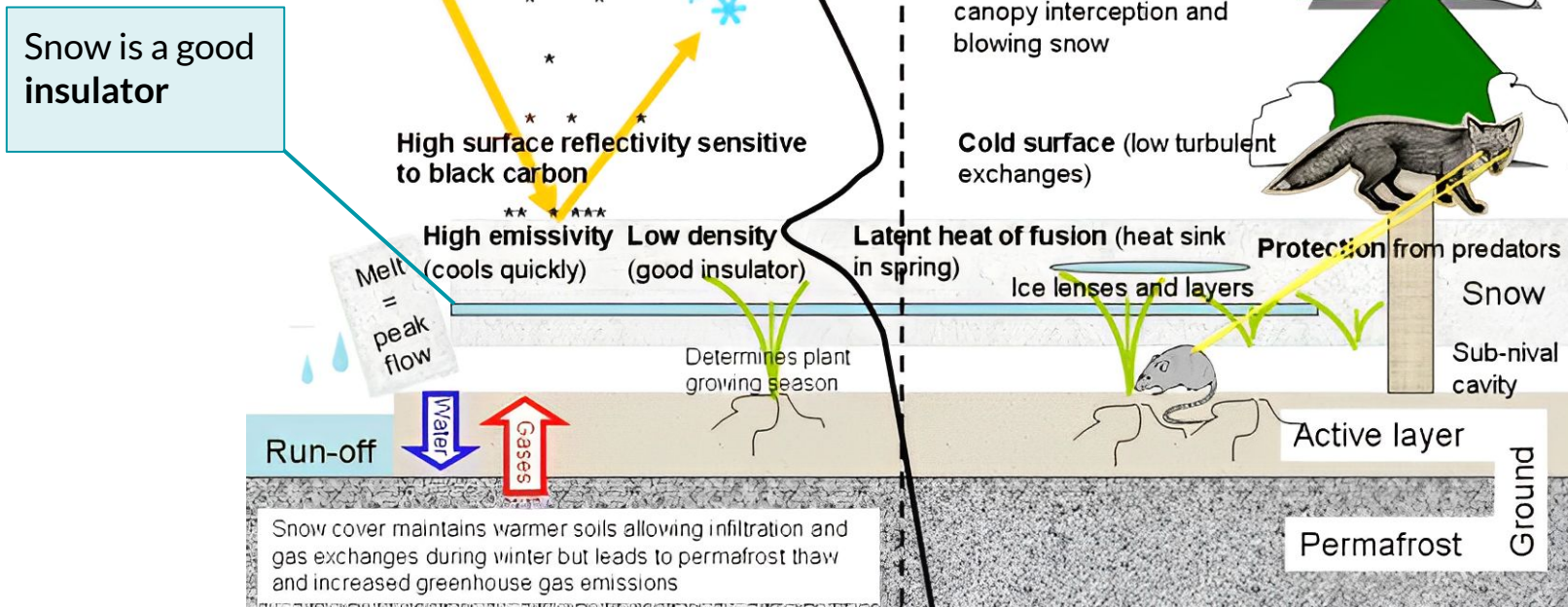
Illustration: Norwegian Polar Institute

Surface energy budget

$$Q_R^* = (1 - \alpha_s)S^\downarrow + L^\downarrow + \epsilon_s\sigma T_s^4$$

Q^* : net radiation

SNOW COVER – an important component of climate, hydrological and ecological systems

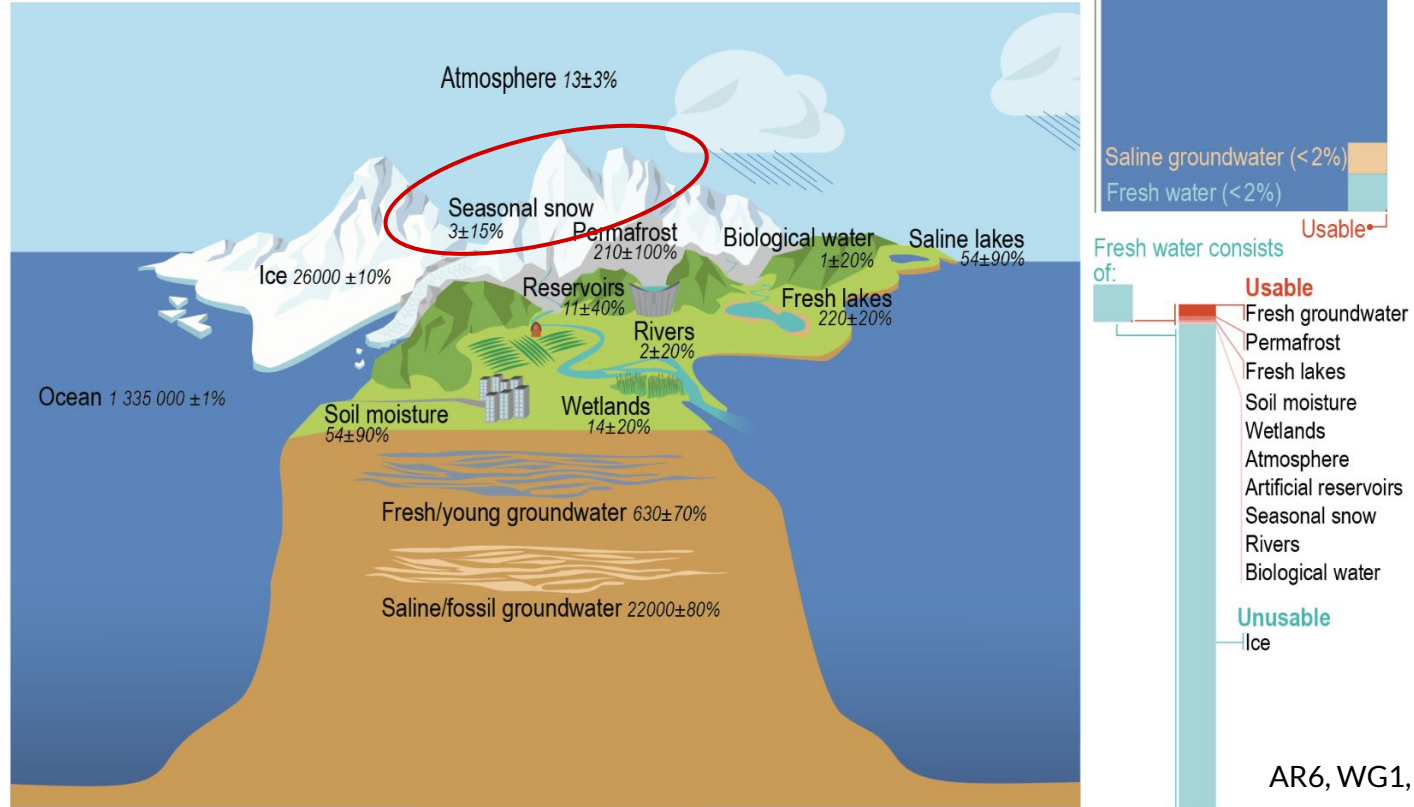


Snow as a part of the terrestrial water storage

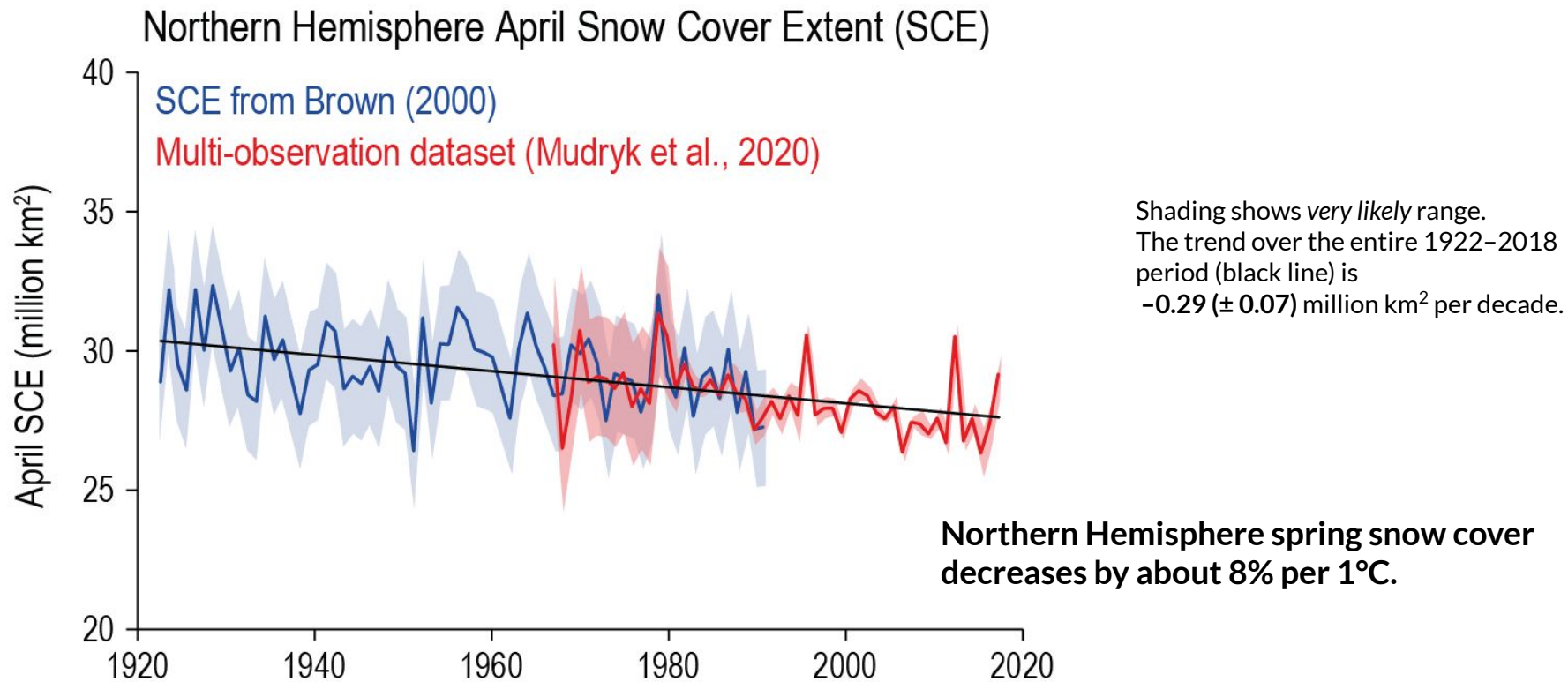
(a) Water stores

Units in thousands of km³

Total water on Earth
(1 380 000 thousand km³)

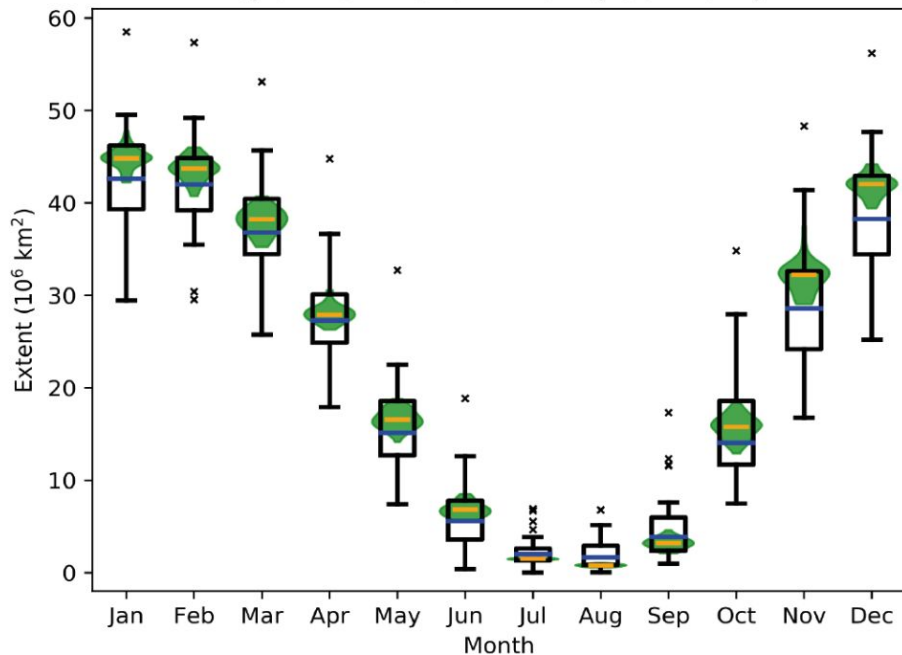


Seasonal snow cover

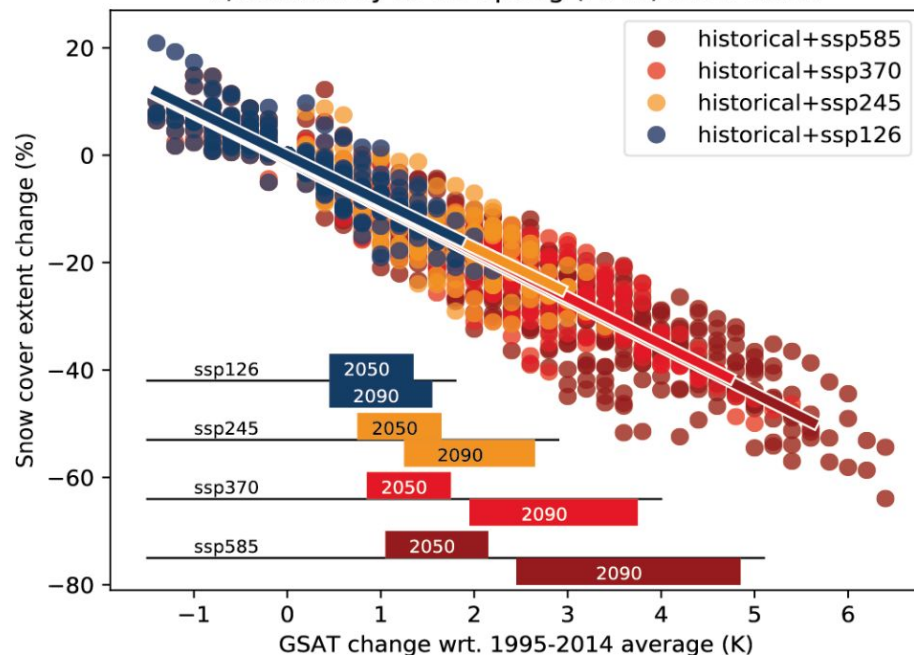


Evaluation of Seasonal Snow in Climate Models

a) CMIP6 NH Snow Extent (1981-2014)



b) Sensitivity of NH spring (MAM) snow cover



How do we observe/measure snow?





*Pierre-Marie Lefeuivre and Simon Filhol
in action in Ny-Ålesund.
Photo: Trine Lise Sviggum Helgerud, NPI*

Read about recent developments on observation systems in the Arctic done by UiO researchers
<https://www.titan.uio.no/innovasjon/2023/spesialbygde-verstasjoner-takler-arktisk-sprengkul.html>

Automated snow monitoring

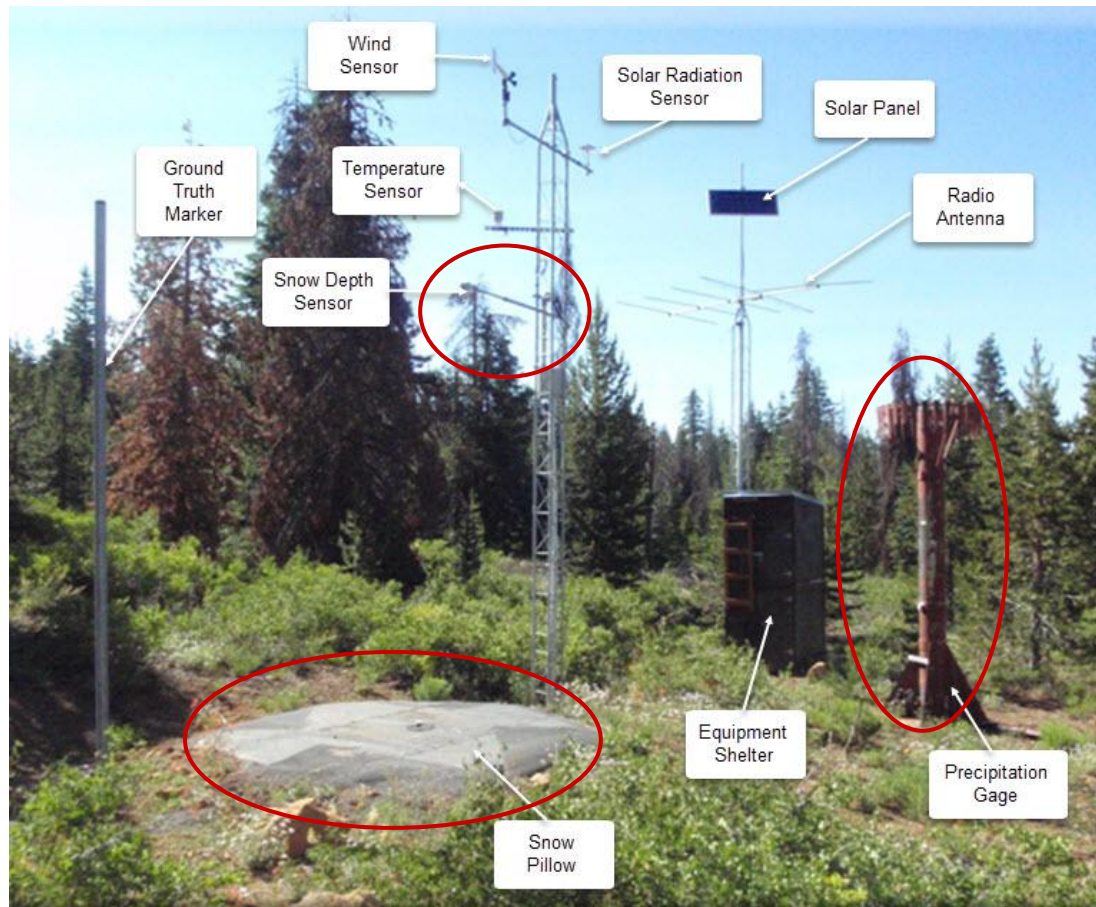
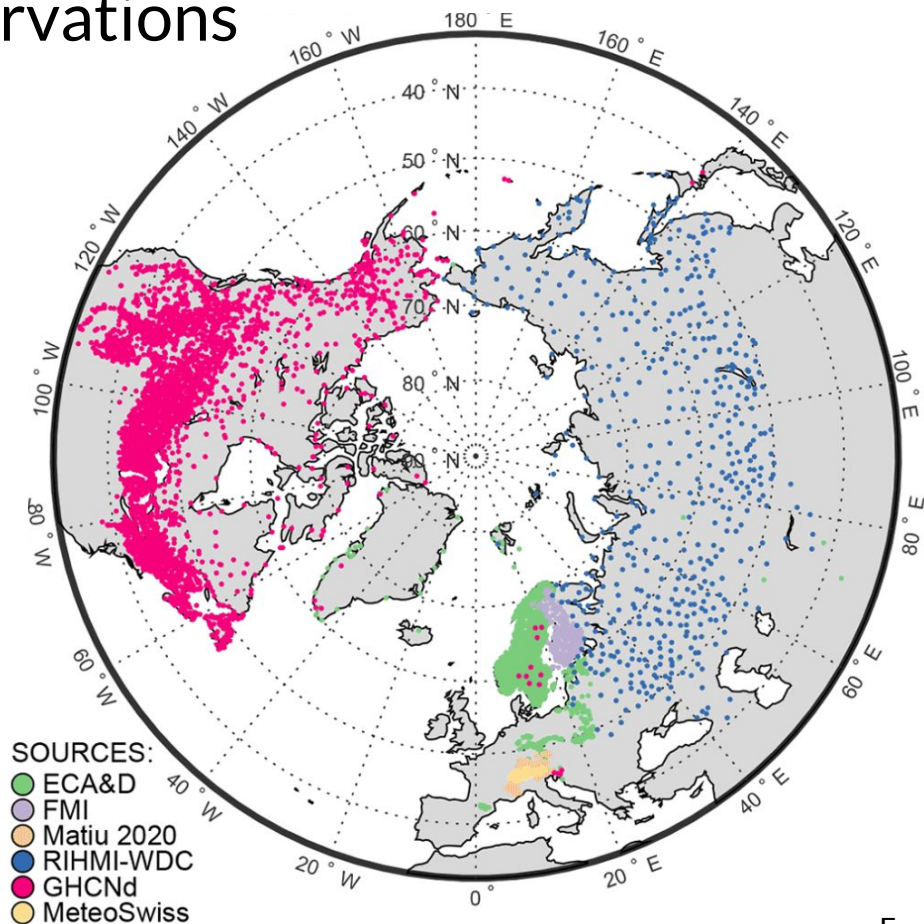


Photo from nrcc.usda.gov

In-situ snow observations



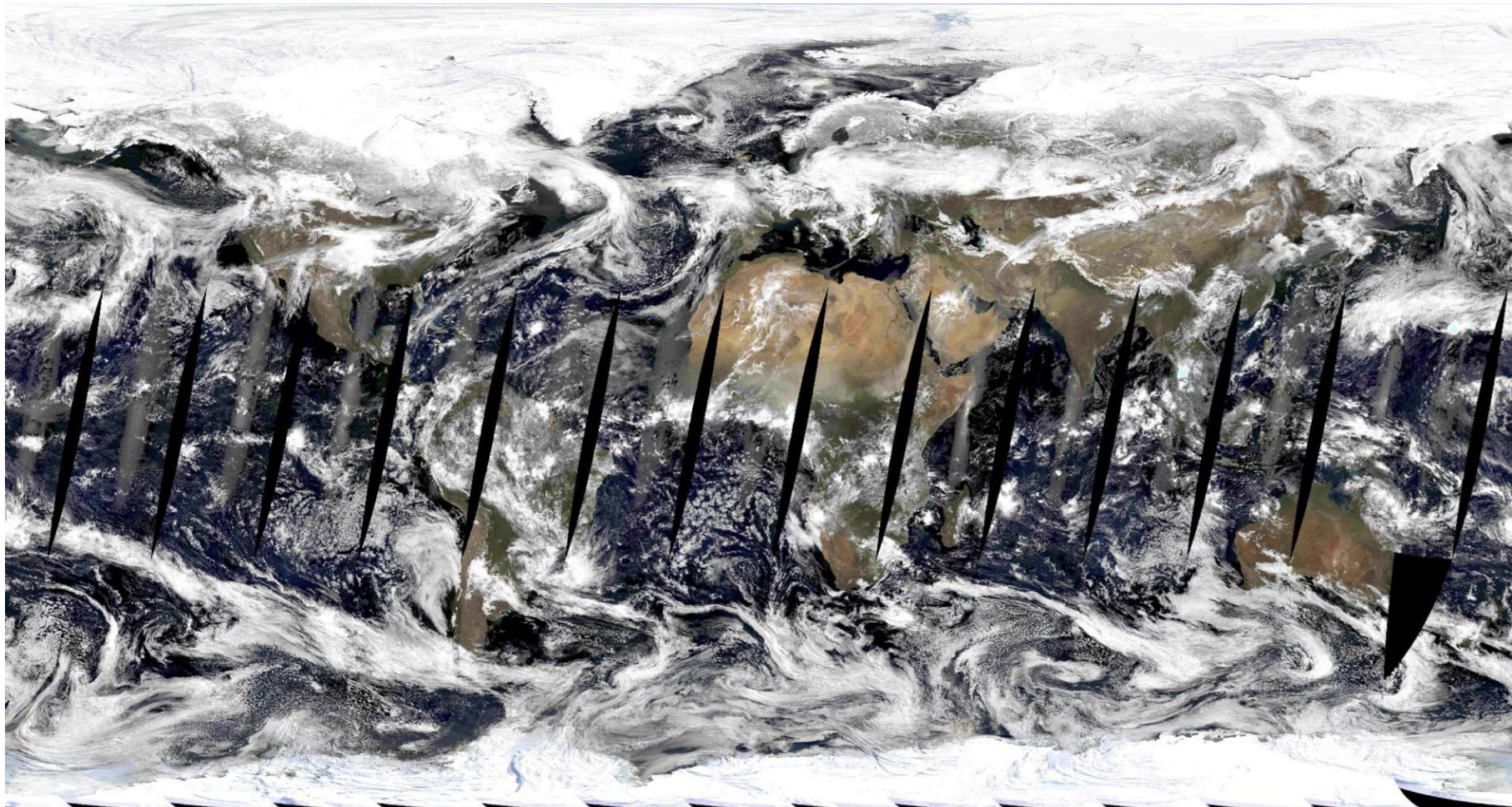


Figure: 01-04-2018 MOD09CMG 0.05° surface reflectance (NASA&USGS) TC RGB=(R,G,B)

© Aalstad

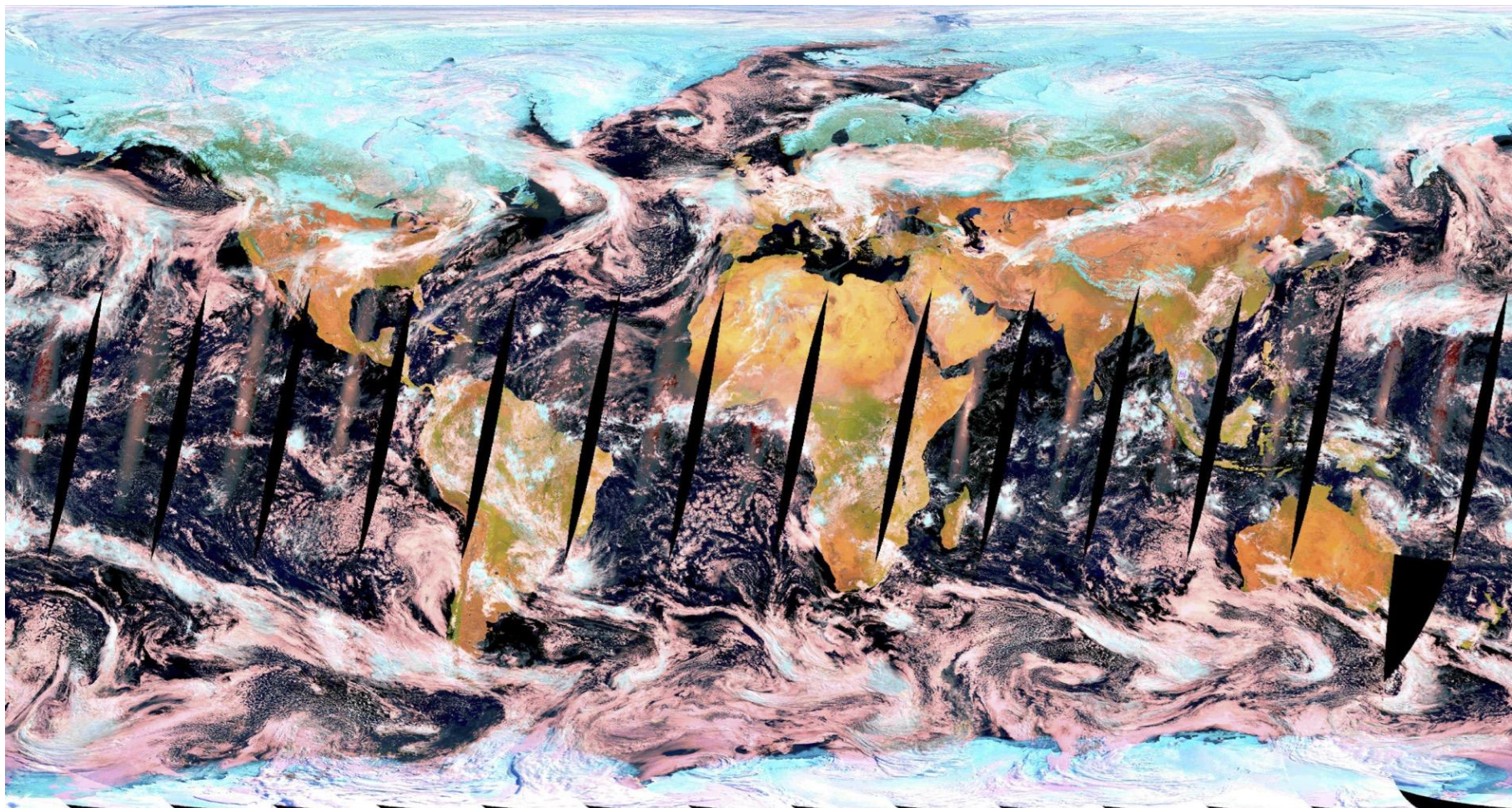


Figure: 01-04-2018 MOD09CMG 0.05° surface reflectance (NASA&USGS) FC RGB=(SWIR1,NIR,R) © Aalstad

Normalized difference-based retrieval algorithms

Normalized difference snow index (NDSI):

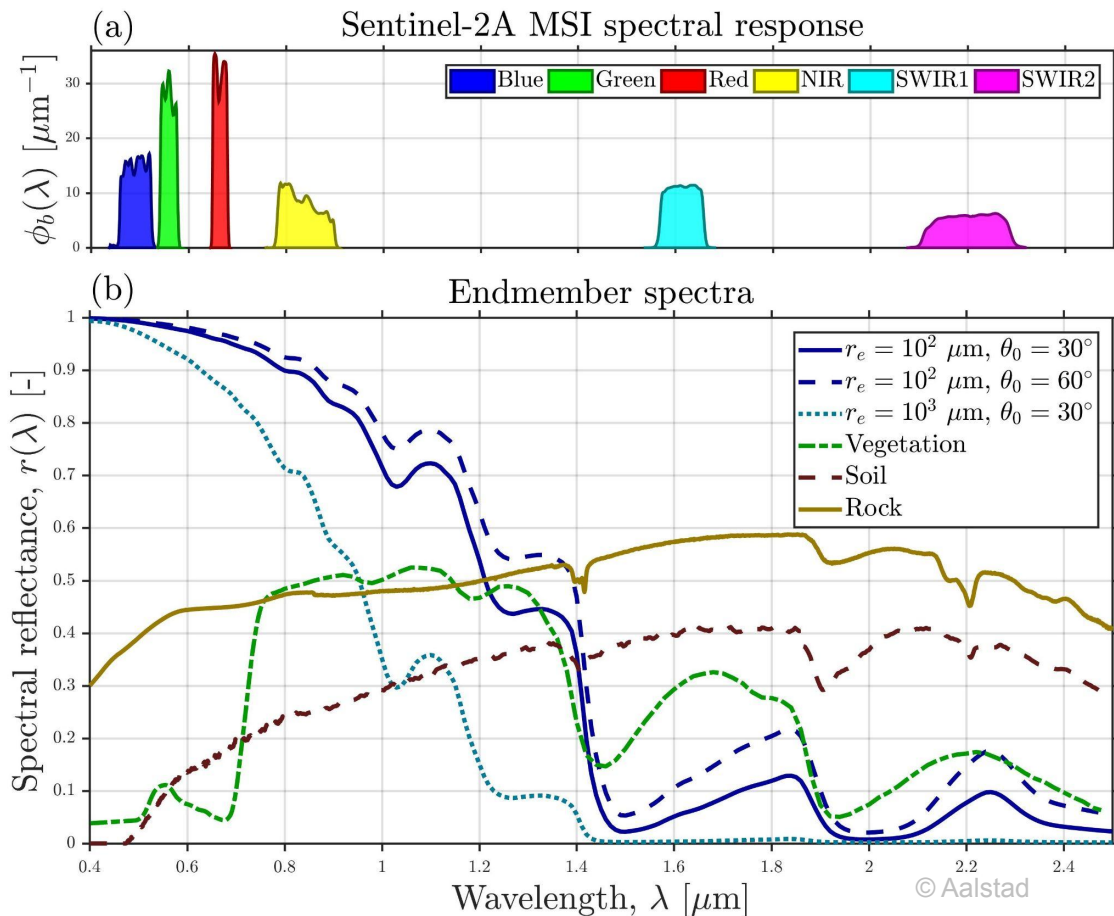
$$\text{NDSI} = \frac{r_{\text{Green}} - r_{\text{SWIR1}}}{r_{\text{Green}} + r_{\text{SWIR1}}}$$



Fractional snow-covered area (FSCA) through linear NDSI regression:

$$\text{FSCA} = \beta_1 \text{NDSI} + \beta_0$$

Aalstad et al. (2020), DOI : [10.1016/j.rse.2019.111618](https://doi.org/10.1016/j.rse.2019.111618)

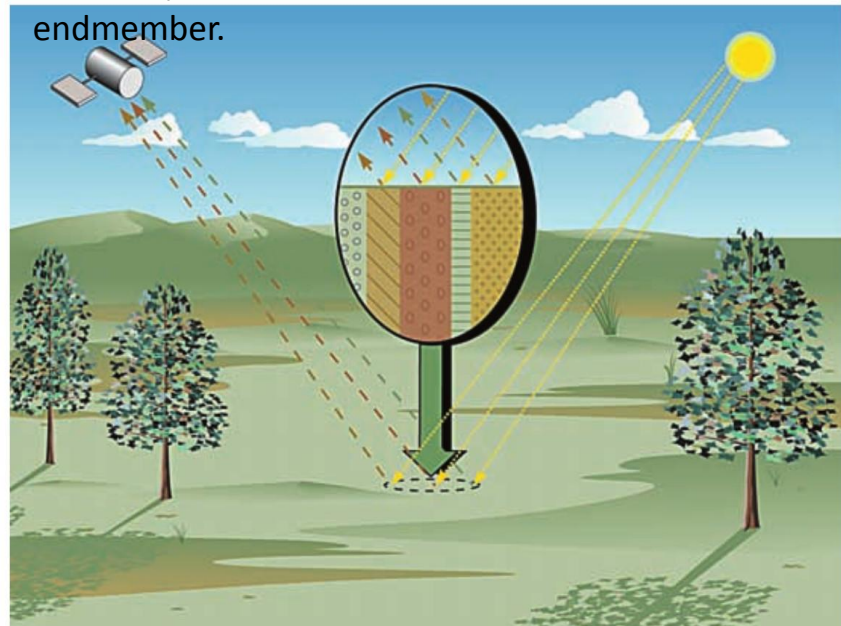


Spectral unmixing retrieval algorithms

Find subpixel endmember abundances $\mathbf{a}$ that satisfy

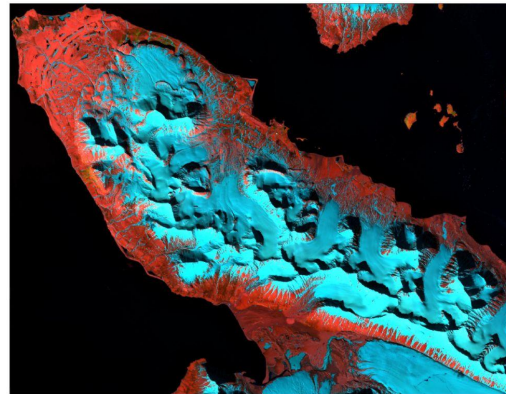
$$\mathbf{a}^* = \operatorname{argmin}_{\mathbf{a}} |\mathbf{M} \cdot \mathbf{a} - \mathbf{r}|^2$$

subject to $\mathbf{a} \geq 0$ and $\sum(\mathbf{a})=1$, where $\mathbf{r}$ is the observed reflectance and $\mathbf{M}$ contains the modeled/theoretical reflectances for each endmember.

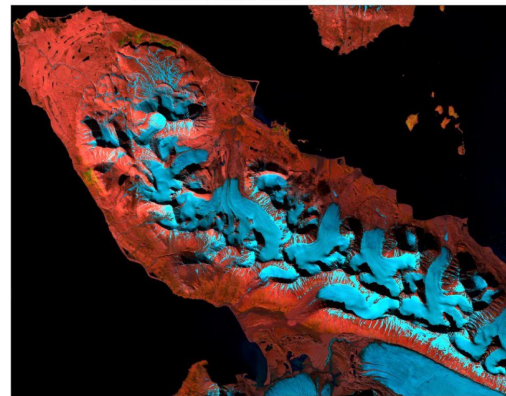


Unmixing “mixed pixels” (Keshava and Mustard, 2002)

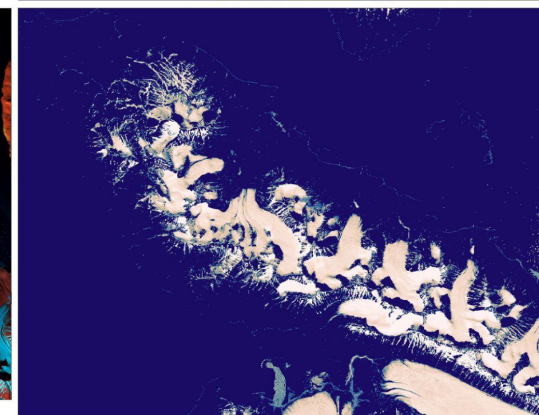
Sentinel-2 False Color 11-Jun-2018



Sentinel-2 False Color 03-Jul-2018

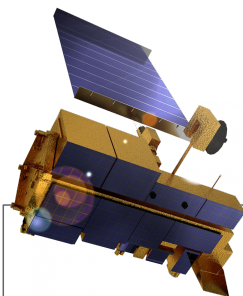
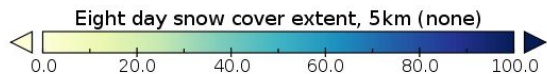
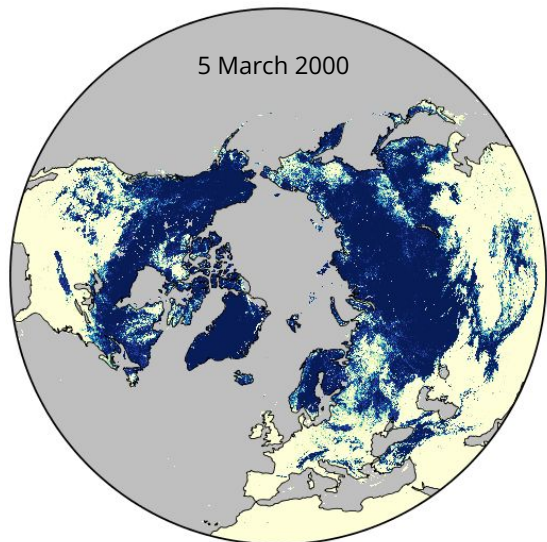


fSCA [black=0, white=1]

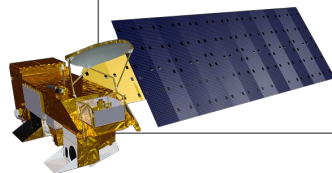


Examples using Sentinel-2 around Ny-Ålesund. Left: SWIR false color images; Right: fSCA. Adapted from Aalstad et al. (2020).

The MODIS Era (MODerate resolution Imaging Spectroradiometer)



Terra

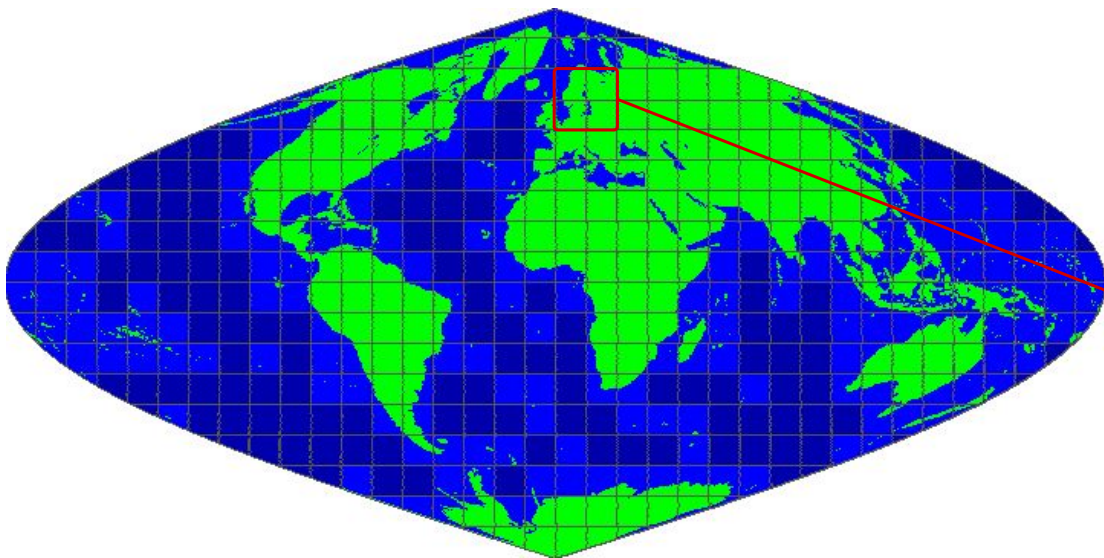


Aqua

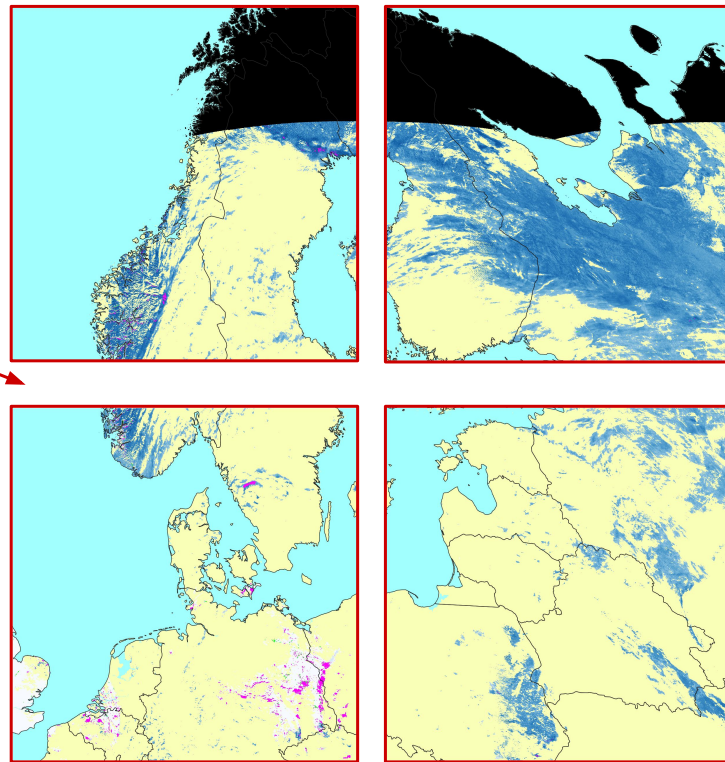
Available global MODIS snow products

MODIS ID	Temporal resolution	Spatial resolution	Temporal coverage
MOD10A1	Daily	500m	24-02-2000
MOD10C1	Daily	0.05 deg	24-02-2000
MOD10A2	8-day	500m	24-02-2000
MOD10C2	8-day	0.05 deg	24-02-2000
MOD10CM	Monthly	0.05 deg	01-03-2000
MYD10A1	Daily	500m	04-07-2002
MYD10C1	Daily	0.05 deg	04-07-2002
MYD10A2	8-day	500m	04-07-2002
MYD10C2	8-day	0.05 deg	04-07-2002
MYD10CM	Monthly	0.05 deg	04-07-2002

Obtaining MODIS snow cover data



MODIS Sinusoidal Tile Grid. Tiles cover $10^\circ \times 10^\circ$ at the equator or approximately 1200 km x 1200 km (Source : NSIDC)



MODIS/Terra tiles over Scandinavia on 28.01.2019

Processing the satellite retrievals

MODIS - NDSI (The Normalized Difference Snow Index)

$$\text{NDSI} = ((\text{band4} - \text{band6}) / (\text{band4} + \text{band6})) \quad (\text{Riggs et al., 2016})$$



fSCA - fractional snow covered area

$$\text{fSCA} = a * \text{NDSI} + b \quad (a, b \text{ are universal constants} - \text{Salamonson and Appel, 2004})$$



Cloud gap filling (Hall et al., 2019)



MODIS bandwidths for selected bands*

Band 4 : 545 - 565 nm (VIS)

Band 6 : 1628 - 1652 nm (SWIR)

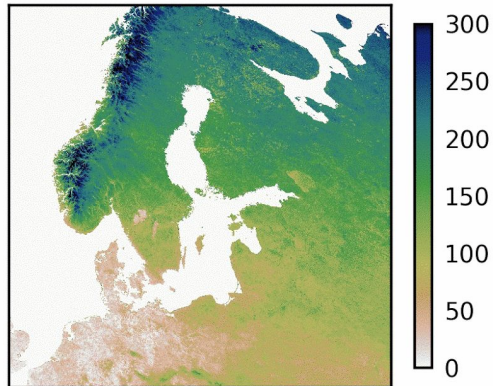
*Snow has very high visible (VIS) reflectance and very low reflectance in the shortwave infrared (SWIR)

Snow metrics (through water year - from 1 Sep to 31 Aug)

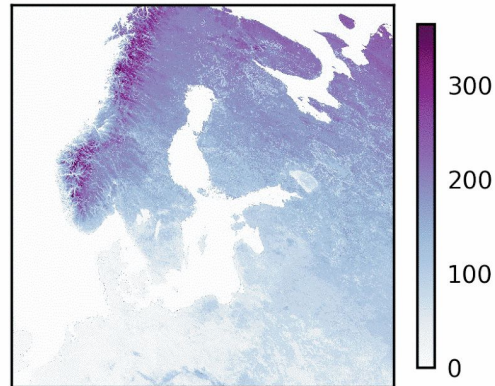
- Snow cover duration (SCD)
- Snow cover extent (SCE)
- Cloud persistence count (CPC)
- First snow day (FSD)
- Last snow day (LSD)
- Consecutive snow cover day (CSC)

2004

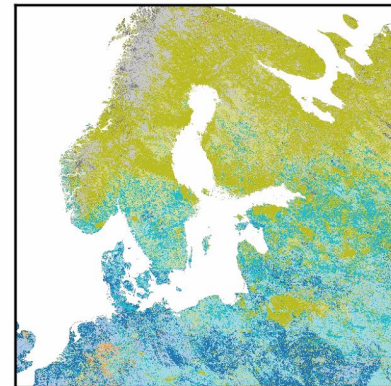
Snow Cover Duration (days)



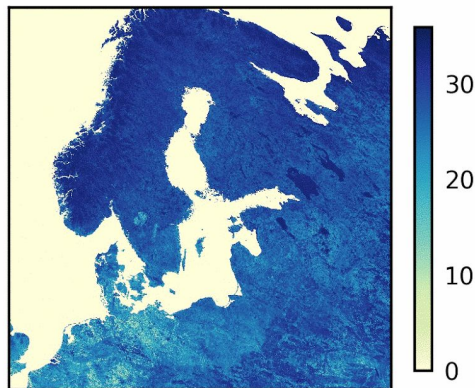
Consecutive Snow Cover (days)



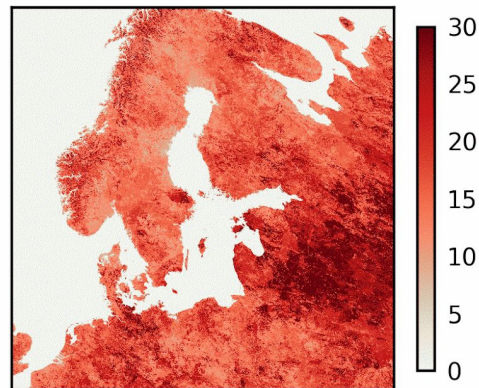
First Snow Day (DOY)



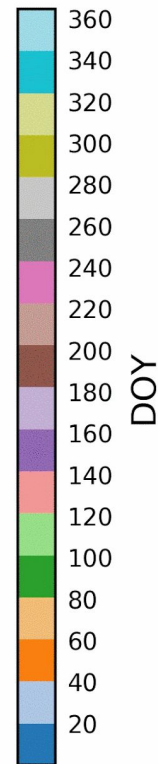
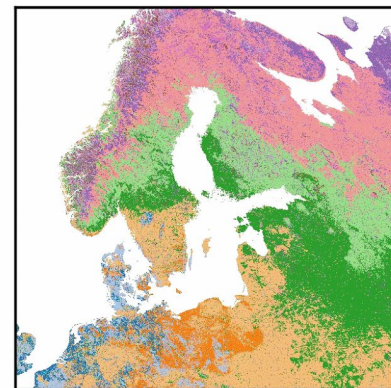
Snow Cover Extent (ha)



Cloud Persistence Count (days)



Last Snow Day (DOY)



Snow cover phenology from combined MODIS snow products (2000 - 2025)

Cloud-free fractional snow-covered area (fSCA)

Daily fSCA maps over Scandinavia at 500 m resolution

Time series of fSCA values over selected sites

Annual snow phenology metrics

Snow cover duration (SCD)

Fractional snow cover duration (fSCD)

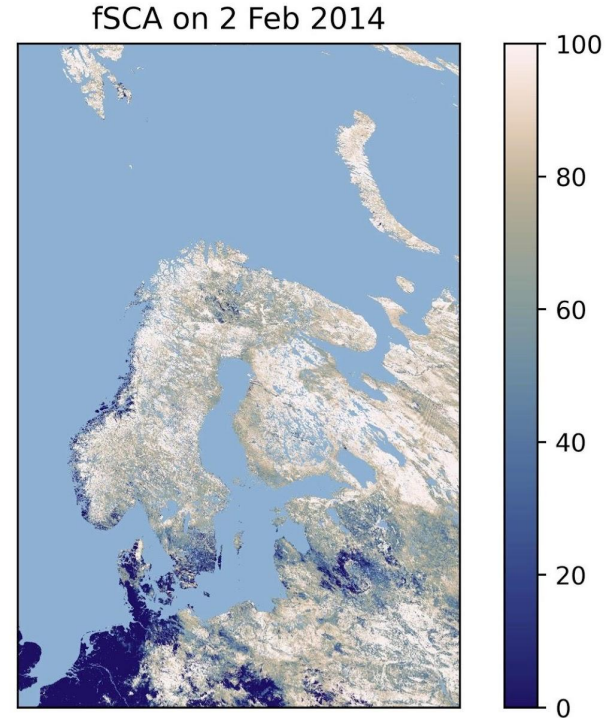
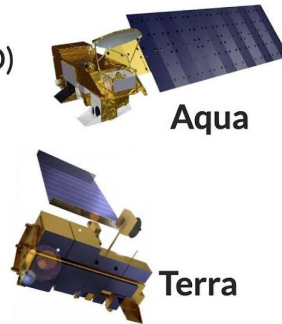
Consecutive snow cover day (CSC)

Cloud persistence count (CPC)

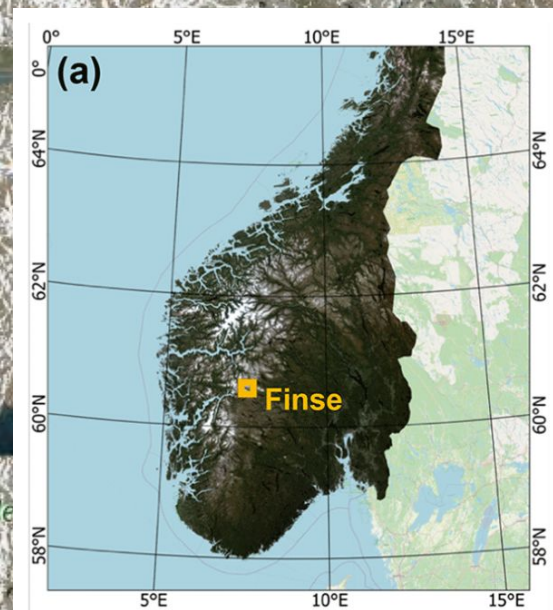
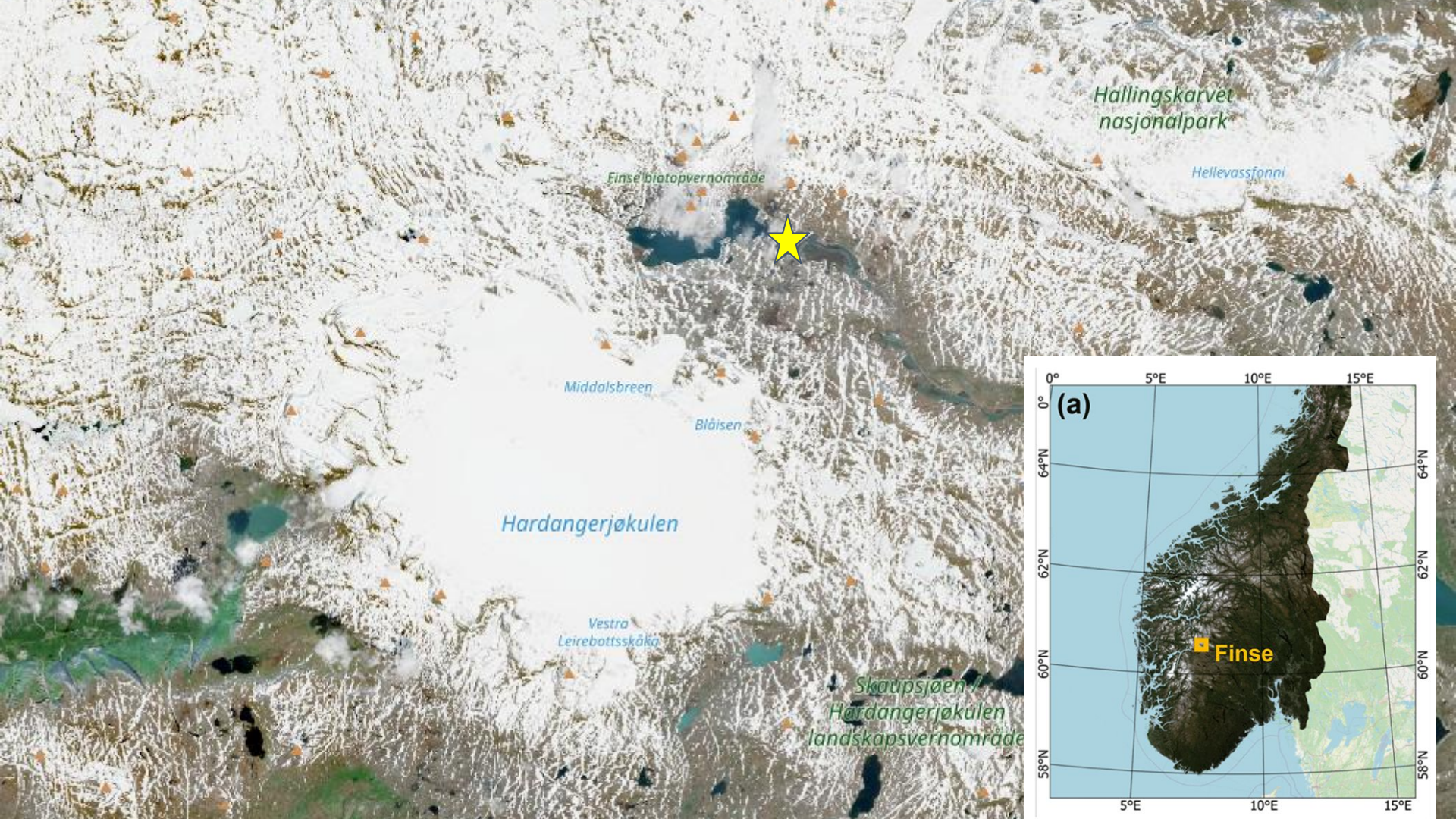
First snow day (FSD)

Last snow day (LSD)

Snow seasonality index (SSI)



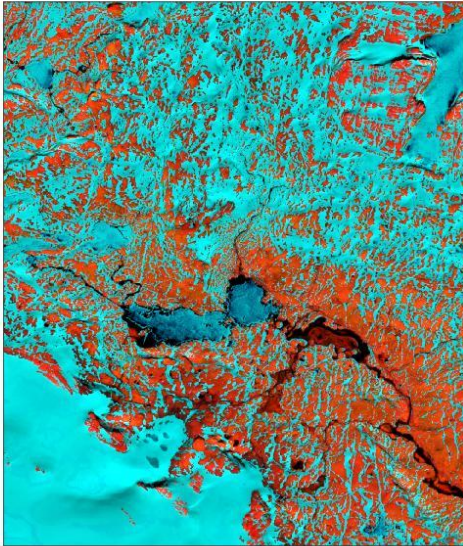
Contact : yeliz.yilmaz@geo.uio.no



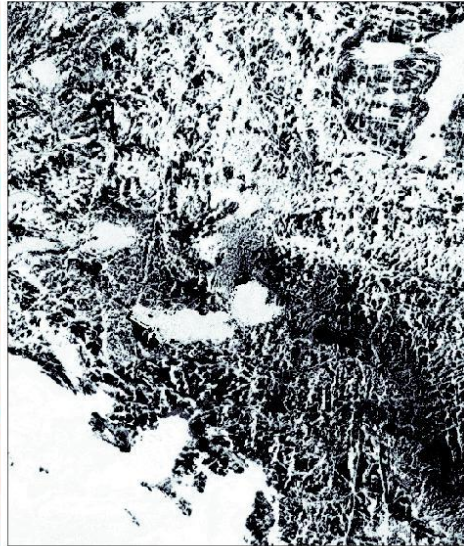
How accurate are the MODIS retrievals?

Sentinel-2

False color image



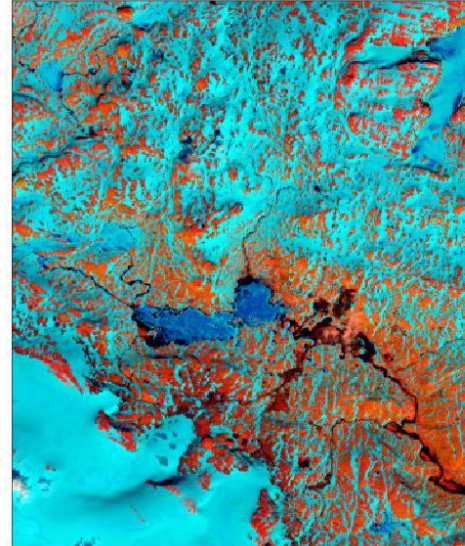
Fractional snow-covered area



08-Jun-2018 11:02:35

Landsat 8 OLI

False color image



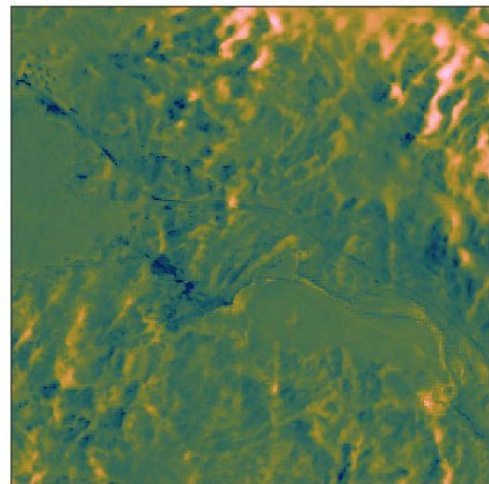
Fractional snow-covered area



06-Jun-2018 10:42:36

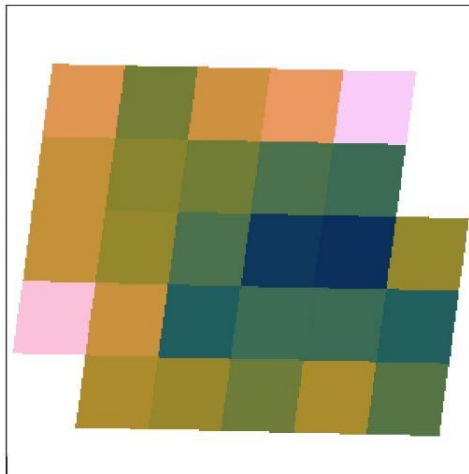
Validation over Finse

LS8+S2 2016-2021 mean

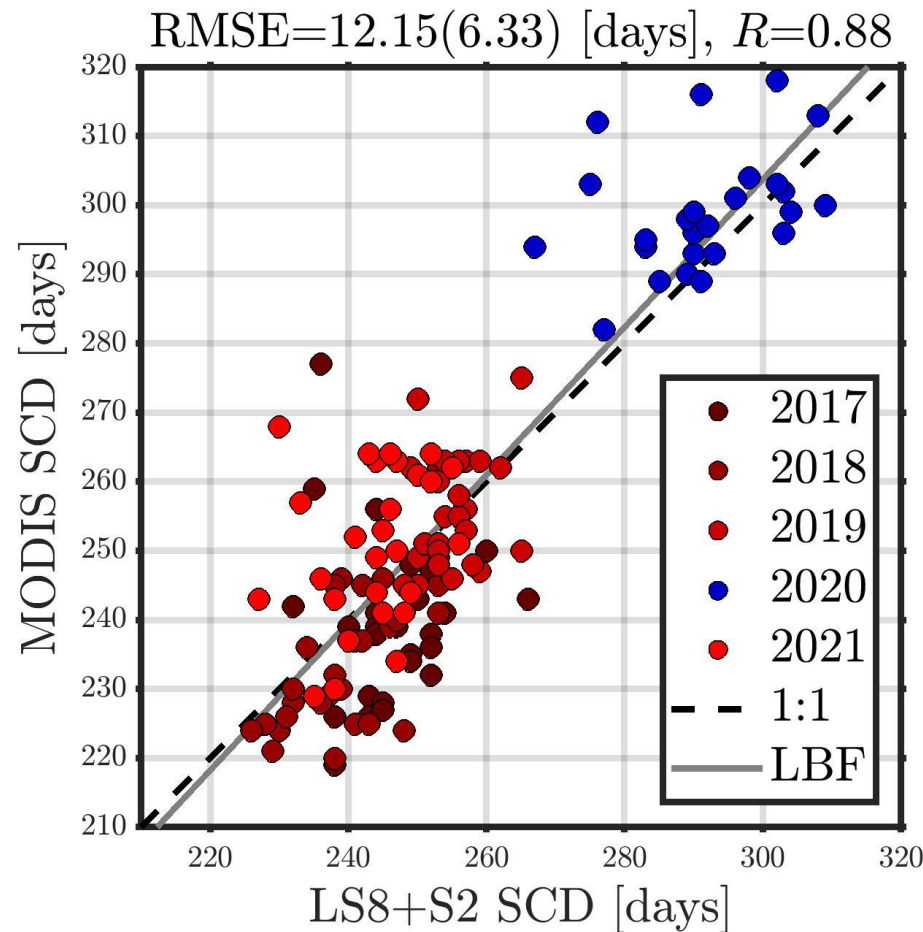


SCD [days]

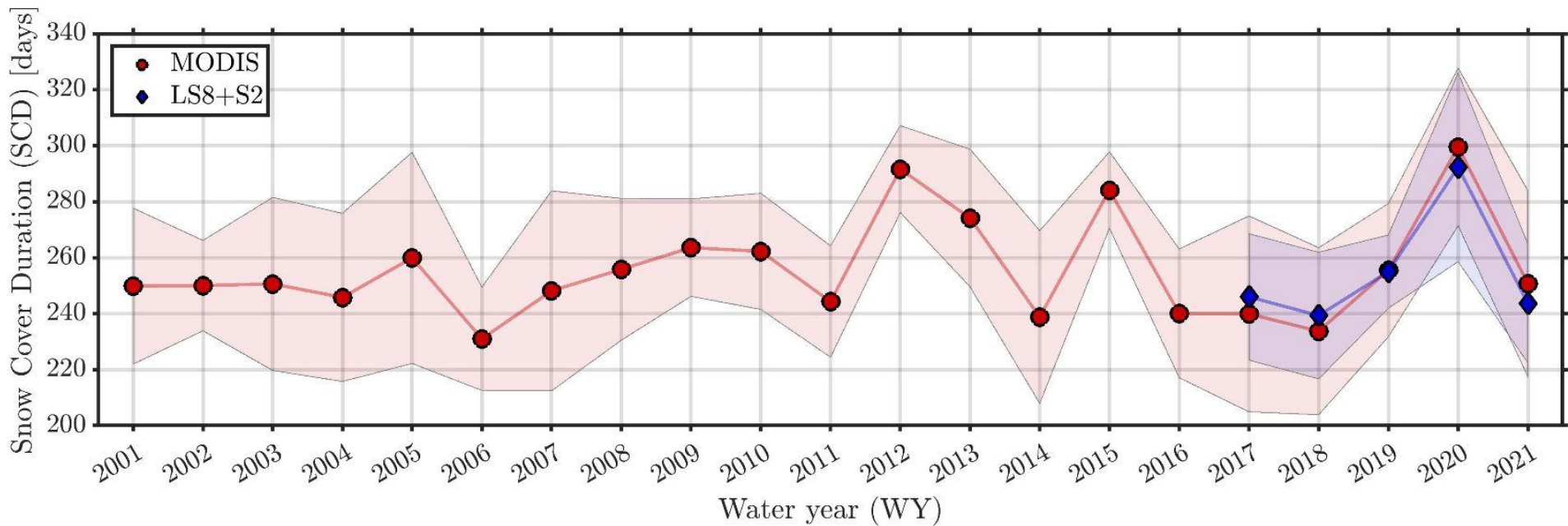
MODIS 2016-2021 mean



SCD [days]



MODIS & high-res. satellite retrievals over Finse





Snow–vegetation–atmosphere interactions in alpine tundra

Norbert Pirk¹, Kristoffer Aalstad¹, Yeliz A. Yilmaz¹, Astrid Vatne¹, Andrea L. Popp^{1,2}, Peter Horvath³, Anders Bryn³, Ane Victoria Vollsnes⁴, Sebastian Westermann¹, Terje Koren Berntsen¹, Frode Stordal¹, and Lena Merete Tallaksen¹

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²Hydrological Research Unit, Swedish Meteorological and Hydrological Institute (SMHI), Norrköping, Sweden

³Natural History Museum, University of Oslo, Oslo, Norway

⁴Department of Biosciences, University of Oslo, Oslo, Norway

Correspondence: Norbert Pirk (norbert.pirk@geo.uio.no)

Received: 30 January 2023 – Discussion started: 6 February 2023

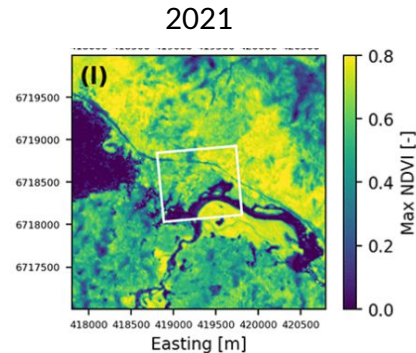
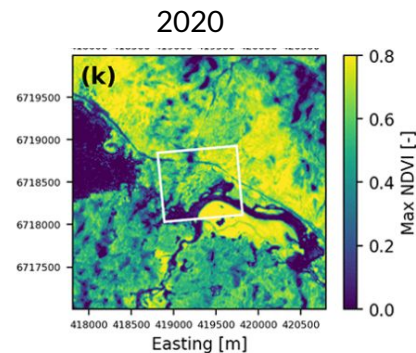
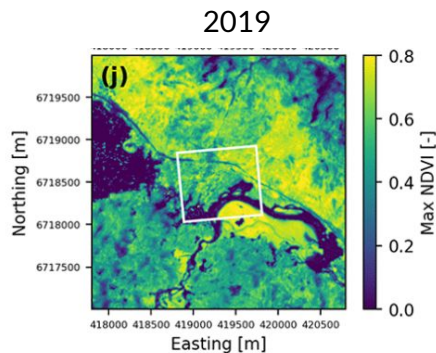
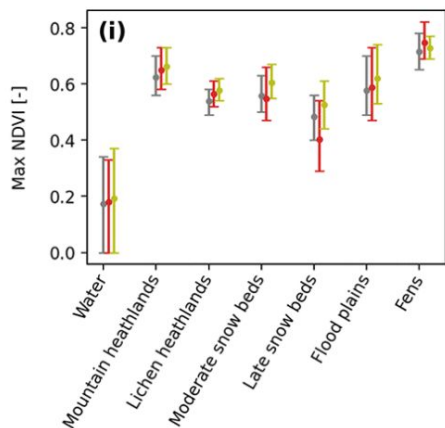
Revised: 14 April 2023 – Accepted: 7 May 2023 – Published: 6 June 2023

Ecosystem-type classification at Finse

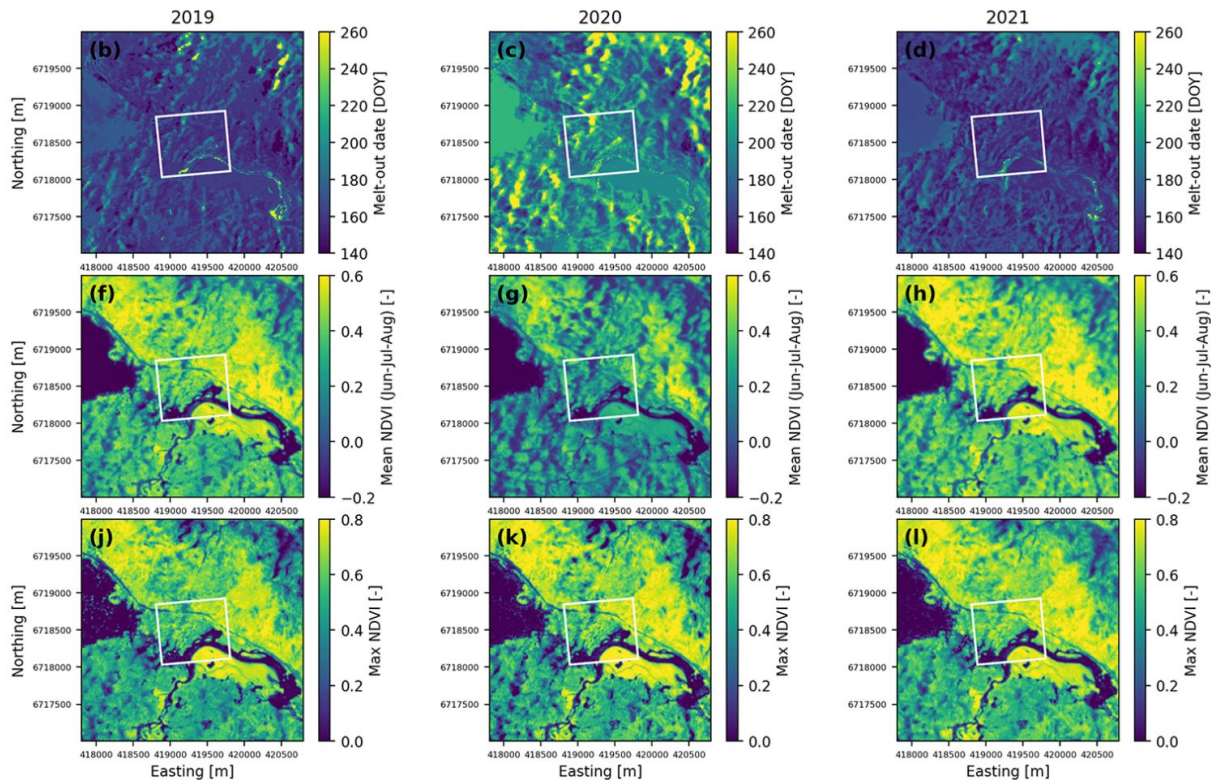
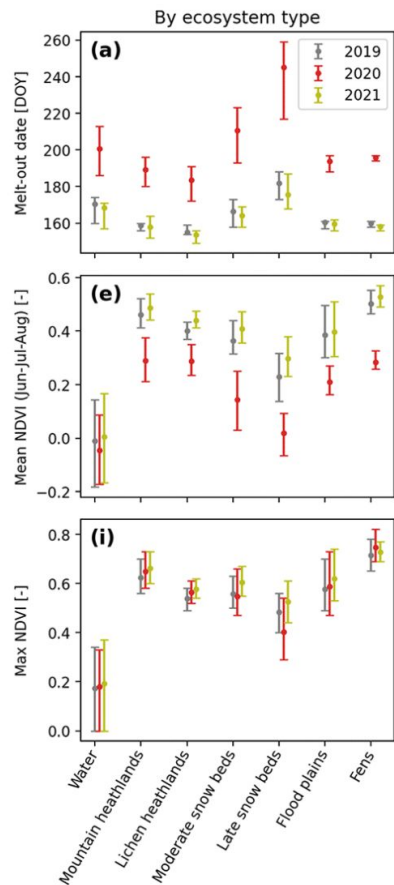
Nature in Norway (NIN) hierarchical mapping system (Halvorsen et al., 2020)

- 43 minor and 13 major types at Finse.
- Reclassified into seven main type categories

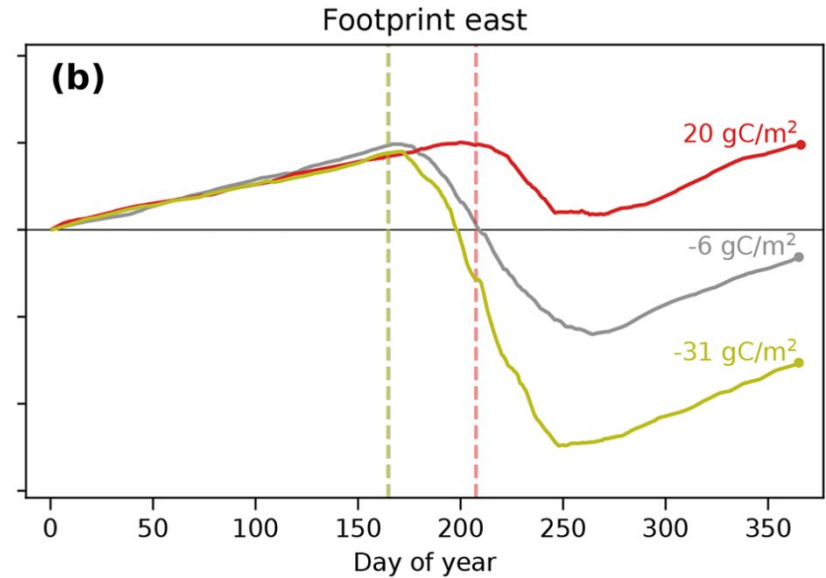
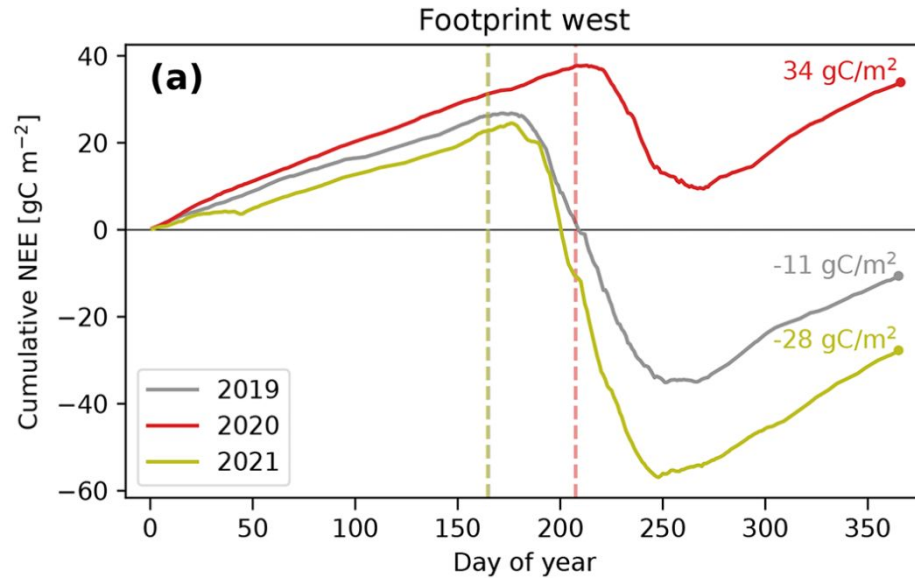
Water
Other
Mountain heathlands
Lichen heathlands
Moderate snow beds
Late snow beds
Flood plains
Fens



Snow - vegetation interactions



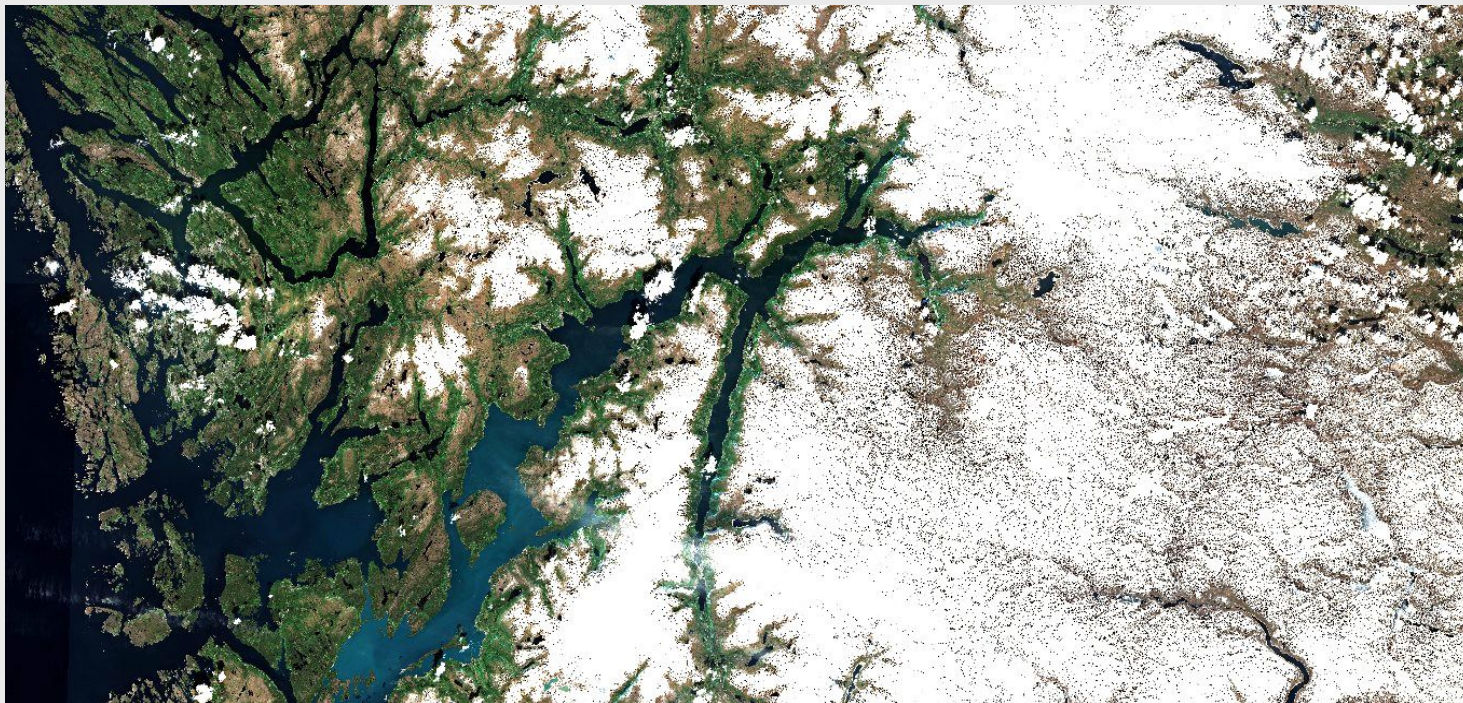
Snow-vegetation-atmosphere interactions



snow melt-out day

2019 2020
2021

	Year	Footprint West	Footprint East
Start date	2019	25 June	18 June
	2020	30 July	18 July
	2021	25 June	20 June
End date	2019	08 September	21 September
	2020	26 September	19 September
	2021	05 September	05 September



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Sentinel-2 natural color image over western Norway
sensed on 02.06.2021 (contains modified Copernicus
Sentinel data [2021] processed by Sentinel Hub)

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<https://www.jobbnorge.no/en/available-jobs/job/308026/phd-research-fellow-in-remote-sensing-of-the-cryosphere>

<https://www.jobbnorge.no/en/available-jobs/job/308033/postdoctoral-research-fellow-position-in-remote-sensing-and-modeling-of-the-cryosphere>

Further Learning Resources

Main source content : <https://www.earthdata.nasa.gov/learn/earth-observation-data-basics/remote-sensing>

https://www.esa.int/SPECIALS/Eduspace_EN/SEMF9R3Z2OF_0.html

Videos : <https://www.youtube.com/watch?v=-loKy-Ey0h8>

<https://www.youtube.com/watch?v=qiqUAIJK4vI>

<https://www.youtube.com/watch?v=MISLC--HNxo>

Exploring the data sets : <https://worldview.earthdata.nasa.gov>

<https://browser.dataspace.copernicus.eu>

Downloading data : <https://earthdata.nasa.gov/> <https://earthexplorer.usgs.gov/>

<https://gportal.jaxa.jp>

earthengine.google.com

<https://www.copernicus.eu/en/accessing-data-where-and-how/copernicus-services-catalogue>

Online Course : ESA Massive Open Online Courses (MOOCs)

<https://eo4society.esa.int/training-education/massive-open-online-courses-moocs>

Canada Centre for Remote Sensing

<https://natural-resources.canada.ca/maps-tools-publications/satellite-elevation-air-photos/educational-resources>